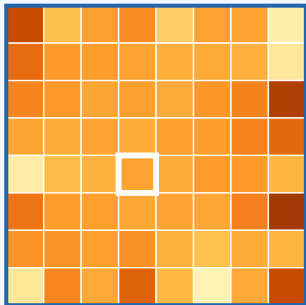


Input observation $\mathbf{y} \in \mathbb{R}^N$

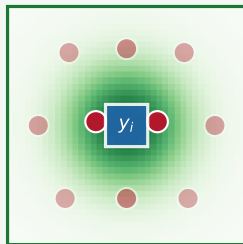
$\leftarrow X_{\text{UTM}} \rightarrow$



Sentinel-2 UTM/MGRS
10 m pixels

build
 $\mathbf{A}$

Sparse operator $\mathbf{A} \in \mathbb{R}^{N \times M}$



$$A_{ij} = K(d_{ij}; \sigma_{\text{PSF}}) \sum_k K(d_{ik}; \sigma_{\text{PSF}})$$



HEALPix cell $\mathbf{x}_j$



UTM pixel y_i

$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}$ · Gaussian PSF (FWHM=12.6 m) Tikhonov prior $\mathcal{R}(\mathbf{x}) = \|\mathbf{x}\|_2^2$

solve

Regularized inversion

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}}$$

$$\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \mathcal{R}(\mathbf{x})$$

Normal equations (SPD):

$$(\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I}) \hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{y}$$

Conjugate Gradient solver

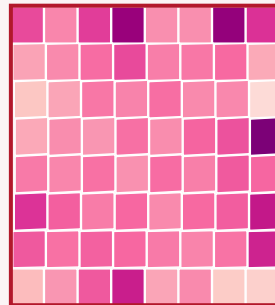
$\mathcal{O}(N)$ per iteration

(sparse $\mathbf{A} \rightarrow$ symmetric SPD)

output
 $\hat{\mathbf{x}}$

HEALPix field $\hat{\mathbf{x}} \in \mathbb{R}^M$

$\leftarrow \phi_{\text{lon}} \rightarrow$



Equal-area cells · $N_{\text{side}} = 2^{20} \cdot \approx 6.3 \text{ m}$

optional $\mathbf{y} = \mathbf{A}\hat{\mathbf{x}}$ (round-trip validation)